

Sugarcane model intercomparison: Structural differences and uncertainties under current and potential future climates[☆]



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ABSTRACT

Sugarcane is one of the world's main carbohydrates sources. We analysed the APSIM-Sugar (AS) and DSSAT/CANEGRO (DC) models to determine their structural differences, and how these differences affect their predictions of crop growth and production. The AS model under predicted yield at the hotter sites, because the algorithm for computing the degree-days is based in only one upper cardinal temperature. The models did not accurately predict canopy and stalk development through time using growth parameters values developed from observed data, in combination with previously determined RUE for the cultivars. In response to elevated CO₂, both predicted higher yields, although AS showed higher sensitivity to CO₂ concentration, rainfall and temperature than DC. The Mean of simulations from both models produced better estimations than predictions from either model individually. Thus, applying the two models (in their current form) is likely to give the more accurate predictions than focusing on one model alone.

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1. Introduction

Decision making and planning in agriculture increasingly utilizes crop models for a variety of tasks, such as yield forecasting in response to either variable climate conditions and/or the impacts of management changes. Most of the crop-growth simulation models used for these tasks are mechanistic and attempt to explain the relationship between parameters and simulated variables and mechanisms of the described processes (Palosuo et al., 2011). Crop models can also be used for generalizing experimental results for broader scales and conditions, compare management strategies, and predict plant growth and production. Such generalizations require robust models that can be evaluated under different environmental conditions to prove their robustness and allow for algorithm improvements when necessary.

Mechanistic crop models are only approximations of reality; therefore, uncertainty is inherent in their model parameters, model structure, and input data, which leads to uncertainty in their predictions (Wallach et al., 2012). One potential approach for reducing

uncertainty that was recently proposed by the Agricultural Modeling Improvement and Intercomparison Project (AgMIP) (Rosenzweig et al., 2013) is to apply multiple models and use the mean or median of the ensemble as the predictors (Asseng et al., 2013). A more traditional approach is to ensure that models are thoroughly tested in the region in which they are being applied.

Brazil is one of the major sugarcane (*Saccharum* spp.) producers in the world. Sugarcane is an important crop for mitigating climate change as a major source for energy and food supplies (Goldemberg, 2007; Brumbley et al., 2008). Therefore, it is valuable to have accurate models for simulating growth, development, and climate change impacts for such crop.

Worldwide, there have been several models developed for sugarcane crop simulations: AUSCANE (Jones et al., 1989), DSSAT/CANEGRO (Jones et al., 2003; Inman-Bamber, 1991; Singels et al., 2008), QCANE (Liu and Kingston, 1995), APSIM-Sugar (Holzworth et al., 2014; Keating et al., 1999, 2003), MOSICAS (Martín, 2003), and CASUPRO (Villegas et al., 2005); however, only two of these models (APSIM-Sugar and DSSAT/CANEGRO) are widely available and supported. The number of sugarcane models is small in comparison to other important crops (e.g., Asseng et al., 2013, who compared 28 wheat models), so further testing and development of these models is clearly important.

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In this paper, we analyzed the APSIM-Sugar (AS) and DSSAT/CANEGRO (DC) models to determine their structural differences and how these differences affect their predictions of crop growth and production. Following the analysis of their structural differences, we 1) assessed the relative skill of the two models in simulating sugarcane production in Brazil; 2) verified if the average of the predictions from the two models gave more accurate predictions of sugarcane production in Brazil than predictions from each model separately; and 3) compared the predictions of sugarcane production under the primary changing climatic variables through a sensitivity analysis.

2. Materials and methods

2.1. Experiment description

Crop models were parameterized for cultivar RB867515, which accounts for nearly 27% (or 1.7 Mha) of the area of sugarcane cultivation in Brazil in 2012 (PMGCA, 2012), using field data from seven locations throughout Brazil (Fig. 1). Two of these experiments had two water-limitation treatments (rainfed and full irrigated), whereas the five remaining experiments were grown under rainfed treatments. All data were collected on plant crops. These datasets represented five distinct soil types and climates of Brazil (Table 1). All of the experiments received adequate N, P, and K fertilization and regular weed control and were planted using healthy cuttings with 13–15 buds/m². Row spacing varied from 1.4 m to 1.5 m. Sites 1 and 2 had two treatments (irrigated and rainfed), and all of the remaining datasets were from rainfed areas. Except Piracicaba, the irrigated treatments received water by drip or sprinkler irrigation that was scheduled following tensiometer monitoring to maintain the soil layers close to the field capacity down to a depth of at least 0.6 m, which ensured full water supply to the crop. In the Piracicaba Site, the soil water balance was used to manage the irrigation and ensure that crops were not exposed to water stress throughout the growing cycle. Irrigation was triggered every time the soil moisture reached 80% of the available soil water.

For Sites 1 and 2, detailed crop growth variables, including the green leaf area index (LAI), stalk population, stalk and aerial dry mass and number of green leaves, were measured at 4–5 week intervals over the cycle. At Site 3, the LAI, stalk population, sucrose concentration, and stalk mass were collected three to seven times at different intervals. For Site 4, the stalk population, stalk mass and stalk height were measured three times during the crop cycle, whereas the sucrose dry mass was

measured 9 times from the mid-season through to harvest. For Sites 5 and 6, the stalk population, stalk mass and stalk height were measured once during the cycle, whereas the sucrose concentration was measured 13 times from the mid-season through harvest. For Site 7, five samples of the LAI, stalk population, sucrose concentration, and stalk mass were measured at regular intervals.

2.2. Model setup and input variables

The soil water dynamics in both of the models were configured in this study with a simple tipping bucket approach and algorithms for the redistribution of water derived from CERES (Jones and Kiniry, 1986). The water characteristics of the soil were specified in terms of the lower limit, drained upper limit and saturated volumetric water contents, and water movement was described using separate algorithms for saturated or unsaturated flow. Because the soil water parameters were not measured in some of the experiments (Olimpia, Colina, and Aparecida do Taboado), the values of water content at –10 kPa (drained upper water limit, DUL), at –1500 kPa (lower water limit, LL) and saturation (saturation water limit, SWL) were estimated using pedotransfer functions (PTF; Tomasella et al., 2000). The input data for the PTF were obtained from Ronaldo Rezende (pers. comm.), Tasso (2007), Andrade et al. (2012), Silva (2007), and Santos (2008). A retention curve was measured for the Piracicaba experiment in which undisturbed soil samples were saturated with distilled water and then the water contents were measured at matric potentials of –0.98, –1.96, –3.92, –9.80, –29.42, –49.03, –98.07, –490.33 and –1471 kPa.

The AS model has the capability to address the redistribution of solutes, such as nitrate and urea; however, this feature was not evaluated in this study because nitrogen stress was avoided in the experiments through adequate N fertilization, as nitrogen stress factor for photosynthesis showed no limitations in the simulations. Surface residue and crop cover reduces soil evaporation in the AS model, but not the DC model. Hence, we set AS model for simulating the crop growth in bare soil, which matches with DC and is a reasonable representation of the soil management for sugarcane plant crops.

The DC model required the input of a root exploration factor for each soil layer, which was a relative variable ranging from 1 (a soil most favorable for root growth) to near 0 (soil unfavorable for roots). Because the distribution of sugarcane root length was similar to an exponential pattern (Ball-Coelho et al., 1992; Laclau and Laclau, 2009), the values were estimated based on the approach proposed by Jones and Ritchie (1991) using the exponential geotropism constant, which is equal to 2.

The AS soil module used an “xf” parameter to govern the relative speed of the root front through a soil layer. This parameter was set at 1 at all sites except those in

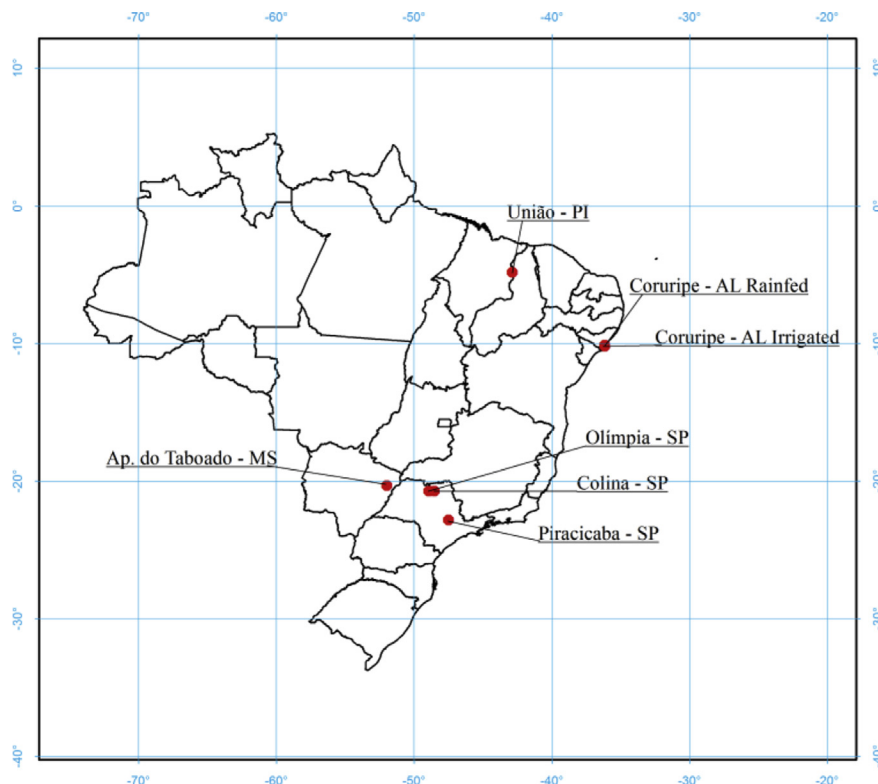


Fig. 1. Location in Brazil of field experiments using cultivar RB867515.

Table 1
Sources of experimental data used, soil and climate characteristics of each site.

Site	Location	Planting and harvest dates	Climate ^a	Water management regime	Soil type	Root depth (cm)
1	União/PI, 4°51'S, 42°52'W, 68 m asml	9/29/2007 and 6/16/2008	27 °C, 1500 mm, Aw	Irrigated and rainfed	Plintossolo (Oxisol) ^b	125
2	Coruripe/AL, 10°07'S, 36°10'W, 16 m asml	8/11/2007 and 11/15/2008	24.4 °C, 1400 mm, As'	Partial Irrigation	Argissolo Amarelo Frágipânico (Fragiudult)	40
3	Coruripe/AL, 10°07'S, 36°10'W, 16 m asml	8/16/2005 and 9/15/2006	21.6 °C, 1400 mm, As'	Rainfed	Argissolo Amarelo Frágipânico (Fragiudult)	40
4	Aparecida do Taboado/MS, 20°05'S, 51°18'W, 335 m asml	7/1/2006 and 9/8/2007	23.5 °C, 1560 mm, Aw	Rainfed	Latossolo Vermelho-Amarelo (Typic Hapludox) ^b	400
5	Colina/SP, 20°25'S, 48°19'W, 590 m asml	2/10/2004 and 12/1/2005	22.8 °C, 1363 mm, Cwa	Rainfed	Latossolo Vermelho-Amarelo (Typic Hapludox) ^b	400
6	Olimpia/SP, 20°26'S, 48°32'W, 500 m asml	2/10/2004 and 12/1/2005	23.3 °C, 1349 mm, Cwa	Rainfed	Latossolo Vermelho-Amarelo (Typic Hapludox) ^b	400
7	Piracicaba/SP 22°41'S, 47°38'W, 540 m asml	10/16/2012 and 06/15/2013	22.4 °C, 1280 mm, Cwa	Irrigated	Argissolo Vermelho (Typic Hapludults) ^b	400

^a Respectively: mean annual temperature, annual total rainfall, Koeppen Classification.

^b Soil Classification by Brazilian Soil Classification System (Embrapa, 1999) and their nearest U.S. Soil Taxonomy equivalent (in brackets).

Coruripe, where α was set at 0.1 from 0.35 m and 0.4 m depth set as 0.1 to represent the physical constraints in the Typic Fragiudults soils. Most of the locations had pH values ranging from 5.5 to 6.5. In Brazil, soils with low pH are usually limed to raise the pH and measurements of pH at a single time may not give a reliable indication of soil acidity. For AS model pH values were set as 7.0 for all layers in order to avoid low pH slowing the process of nitrification (Marin et al., 2013). In the DC implementation we used here the soil N module was not activated.

The weather input data required for both models included the daily solar radiation, maximum and minimum air temperature, and rainfall. The relative humidity and wind speed are not essential but are recommended whenever available. For Sites 1, 2 and 3, the air temperature, wind speed, relative humidity and solar radiation were collected by automatic weather stations positioned in proximity to the experiments. For Site 4, the air temperature and solar radiation were collected by an automatic weather station positioned approximately 15 km from the experiment. A flat terrain minimized the error associated with this procedure. Sites 5 and 6 were approximately 30 km from each other over flat terrain. The maximum and minimum air temperature data were collected daily at the same weather station approximately 15 km from both the experiments, and the wind speed and relative air humidity were not measured for these sites. To avoid the inclusion of another uncertainty source in the simulations, we input only the minimum weather data (solar radiation, air temperature and rainfall) to both models for all of the sites. For the sites at which daily solar radiation was not available, the data were estimated using the Bristow and Campbell method (1984). For all of the datasets, rainfall was recorded at a distance less than 100 m from the experimental fields. For Site 7, all of the required weather data were collected daily from a weather station close to the experimental area. Missing weather data values were estimated through the WGEN algorithm (Richardson and Wright, 1984) and by interpolated estimates from nearby stations, as done by Marin et al. (2012). The AS model can simulate ET by a number of methods, and the DC model can simulate ET by the Penman–Monteith (PM) and Priestley–Taylor methods. The PM was selected in this study for both models.

2.3. Model comparison, crop cultivar coefficient calibration, evaluation, and sensitivity analysis

The primary model processes and structures in the two models are compared in Table 2. The NCo376 cultivar coefficients were available for the general AS and DC releases and assumed to be developed against robust experimental field data. Thus, this cultivar was chosen as the base from which to start the process of calibrating the cultivar coefficients for the seven experiments. Prediction of the experimental data was first made with NCo376 cultivar coefficients to assess how closely the field results could be simulated with this standard cultivar.

Following these initial simulations, values of the crop parameters such as leaf size, partitioning and tiller and leaf size relationship were set in both models based on the observed data. Further, the phenology, canopy structure, and partitioning-related parameters values were set based on observed data from Sites 2, 3 and 7 where these variables were measured. The parameter values were selected to minimize the RMSE between the predicted and measured data for the stalk fresh mass (SFM) and LAI. For both of the models, the calibration process followed a sequence that started by calibrating the phenology, stalk population, canopy cover of irrigated experiments, canopy cover of rainfed experiments, and biomass partition fractions. The stalk height and sucrose were not included in the calibration process.

The response of sugarcane growth and yield to changes in temperature, CO₂ concentration and amount of rainfall was analyzed in both models by changing one parameter at a time from the baseline scenario, which reflected the actual historical

conditions for the experimental site. The temperature of the baseline climate was changed by -3 °C to $+9$ °C in 3 °C steps, the CO₂ concentration was changed from 350 ppm to 750 ppm in 100 ppm steps, and the rainfall of the baseline climate was changed from -30% to $+30\%$. An historical weather series for Piracicaba-SP from 1980 to 2010 was used as the baseline climate, and incremental changes in the temperature and rainfall were set with the “climate control” module in the AS and environmental modifications module available in the DS. Piracicaba-SP was chosen as a reference location for this analysis because it represents the most important Brazilian sugarcane production region. For those analyses, both models used a soil profile with high water retention capacity and rooting depth (4.5 m). Different atmospheric CO₂ concentrations were specified directly in the crop XML file of the AS and environmental modifications module of the DSSAT. A sensitivity analysis was conducted with the same management conditions as the Piracicaba experiment (Site 7, Table 1).

3. Results and discussion

3.1. Model intercomparison with a standard cultivar

Simulations with both the DC and AS models using the standard parameters for the NCo376 cultivar were reasonably close to the experimental data for the RB867515 cultivar at Sites 2, 3 and 6 for SFM (Fig. 2c, d, and g); however, the LAI predictions at Sites 2 and 3 were not close to the experimental data (Fig. 3c and d). The SFM predictions of both models showed major differences from the measured values for Sites 5 and 7 (Fig. 2f and h), although the DC model estimates of LAI were consistent with those observed at Site 7 (Fig. 3h). For site 1, the DC model had better predictions of SFM than the AS model (Fig. 2a and b).

Because of the different temperatures across the different sites (Table 1) and the effects of temperature in both of the models in terms of phenology, tillering and leaf area algorithms, the different responses in Fig. 2a and b between the models was likely to be driven by the climatic differences across the sites, notably different air temperatures. In the DC model, the plant emergence was a function of the temperature and the reference South African cultivar NCo376 required 428 and 203 °C d for the plant and ratoon crops, respectively. In the DC, the peak tiller population for NCo376 was attained at 600 °C d. The AS model also used air temperature to simulate the phenology, tillering and leaf area (Table 2), considering a base temperature of 9 °C, optimum temperature of 32 °C, and maximum temperature of 45 °C, which represents a different approach of DC model. An important difference between the models was in their methods of integrating thermal time. As described in Table 2, the AS model was limited in its ability to accumulate degree-days in hot places because of the use of only one upper cardinal temperature (i.e. a single maximum value) with

Table 2

Comparison of main processes simulated and model features for DSSAT/CANEGRO and APSIM-Sugar.

DSSAT/CANEGRO	APSIM-Sugar
Base temperature and degree days Thermal time ($^{\circ}\text{C d}$) is calculated based on different base temperature for photosynthesis (7°C), emergency, start of elongation and leaf phenology (10°C) and stalk development (16°C). The last one is assumed to be cultivar specific. There is no optimum or maximum temperature for plant phenology and biomass increase, but there is an optimum temperature for plant respiration, as explained below. Thermal time is calculated in a daily step basis an optimum or maximum temperature limitation.	Base temperature is fixed at 9°C for all crop development with an optimum temperature (32°C) and maximum temperature (45°C). These are not cultivar specific, assumed constant for the species. Thermal time ($^{\circ}\text{C d}$) is calculated based on 3-hourly air temperatures interpolated from the daily maximum and minimum temperatures, using a base temperature, optimum, and maximum temperature, as specie parameters.
Emergence and tillering Primary tiller emergence is simulated when a specified accumulated thermal time has accumulated from planting or ratooning. Peak tiller occurs when a cultivar-specific amount of thermal time has accumulated since emergence. The tillering phase lasts from the emergence of primary tillers to the occurrence of peak tiller population, while the stalk elongation phase lasts from the start of stalk elongation up to harvest. The tiller senescence phase commences after peak tiller population is reached and continues until harvest. Shoot cohorts are phased in (and out) to account for the range in shoot age at a given point in time. No distinction is made between plant and ratoon crops apart from the thermal time requirement for emergence of primary shoots. Model also does not take in account the shoot rate elongation up to emergence. This implies that only the required degree-days regulates the emergence date, and not the planting depth.	Sprouting occurs after a lag period, set to 350°C d for plant crops and 100°C d for ratoon crops. Provided the soil water content of the layer is adequate, shoots will elongate towards the soil surface at a rate of $0.8\text{ mm }^{\circ}\text{C}^{-1}\text{ day}^{-1}$, after a thermal time lag of 250°C d. The thermal duration between emergence and beginning of stalk growth is a genotype coefficient in the range $1200\text{--}1800^{\circ}\text{C d}$. Stalk number rises rapidly to a peak during the first 1400°C d from emergence, thereafter declining to reach a stable stalk number. Ratoon crops reaches an earlier peak stalk number than plant crops, with consequently faster early canopy expansion in ratoons. Stalk population does not change in APSIM unless the crop lodges. In the model, to avoid the complexity of simulating the dynamics of tillering in order to predict LAI, the crop is conceived to have a notional constant stalk number throughout growth, and the additional leaf area associated with tillers is captured via a calibrated tillering factor.
Photosynthesis and respiration Radiation use efficiency computed at canopy level calculated as a function of a maximum radiation use efficiency, assumed as a cultivar parameter, air temperature and base temperature for photosynthesis. Growth respiration is assumed a species parameter given by 0.242 t t^{-1}. Maintenance respiration is calculated as an exponential function of temperature. No distinction is made between plant and ratoon crops, and temperature is not considered on RUE calculus.	Intercepted radiation is used to produce daily biomass production using a radiation-use efficiency parameter of 1.80 g MJ^{-1} for plant crops and 1.65 g MJ^{-1} for ratoon crops, assumed as a constant for the species. In the model, RUE computed at canopy level and is reduced if the mean daily temperature falls below 15°C or exceeds 35°C, and becomes zero if the mean temperature reaches 5 or 50°C, respectively. Water stress affects the photosynthesis rate as explained below. Respiration is not directly computed.
Partitioning and biomass accumulation Four above-ground biomass pools are modeled: leaf, cabbage, structural stem, stem sucrose. Daily partitioning of assimilate between roots and aerial parts is simulated as a non-linear function of total biomass. A large fraction is partitioned to roots early in the life of the plant but this decreases rapidly as the plant ages. The maximum fraction of daily biomass increments that is partitioned to aerial parts (in a mature crop) is specified in the cultivar file. A temperature-dependent fraction of aerial dry mass is partitioned to stalk and is limited to a cultivar-specific maximum value when thermal time since emergence exceeds a cultivar-specific threshold.	Four above-ground biomass pools are modeled: leaf, cabbage, structural stem, stem sucrose, and an additional pool for roots that is simulated separately from above-ground production. Between emergence and the beginning of stalk growth, above-ground biomass is partitioned between leaf and cabbage in the ratio 1.7:1. Thereafter, the stem biomass is equal to the sum of structural and sucrose pools, considering a source-sink approach for leaf area development. A stalk growth stress factor is calculated as the most limiting of the water, nitrogen and temperature limitations on photosynthesis, influencing both the onset and rate of assimilate partitioning to sucrose at the expense of structural stem.
Sucrose accumulation Sink capacities for structural growth and sucrose storage are dictated by current growing conditions (temperature, water status), current stalk mass and cultivar characteristics. The sucrose accumulation component of the model is based on a framework of sucrose distribution within stalks as it is affected by temperature and water stress. Sucrose accumulation response to temperature is determined by a species parameter controlling the sucrose partitioning response to temperature, and a cultivar parameter controlling the plant ability to accumulate sucrose. Sucrose partitioning response to water stress is captured through an ecotype parameter.	After a minimum amount of stem biomass has accumulated, the daily biomass partitioned to stem is divided between structural and sucrose pools. If biomass available for partitioning to leaf is in excess of that required to grow the leaf area on that day, as determined by the minimum SLA, 'excess' biomass is partitioned to sucrose. A stalk growth stress factor is calculated as the most limiting of the water, nitrogen and temperature limitations on photosynthesis. This stress factor influences both the onset and rate of assimilate partitioning to sucrose at the expense of structural stem.
Root growth Root growth is expressed in terms of the extension of the rooting depth, as well as the increase of root mass and root length per soil layer. The latter, together with soil water status, determine the potential crop water uptake used to calculate the level of water stress. Root length density assumed that 0.5% of root length density is lost each day. Daily potential increase in rooting depth is calculated assuming a 2.2 mm penetration per unit thermal time for that day, until the maximum rooting depth is reached. Rooting depth is initialized at zero for a plant and is set to the final rooting depth of the preceding crop for ratoon crops.	Root biomass is produced in proportion to the daily aboveground biomass production and affected by water stress. The proportion decreases from a maximum of 0.30 at emergence and asymptotes to 0.20 at flowering. Root biomass is converted to root length via a specific root length. The depth of the root front in plant crops increases by 1.5 cm day^{-1} from emergence (assuming no soil limiting to root front advance), with the maximum depth of rooting set by the user. At harvest, 17% of roots in all the occupied soil layer die.

(continued on next page)

Table 2 (continued)

DSSAT/CANEGRO	APSIM-Sugar
Leaf area index and extinction coefficient Leaf appearance is based on the phyllochron concept using a “broken stick” approach. Two phyllochron values apply to leaves below and above a cultivar-specific threshold, respectively. The senescence of green leaves is based on the assumption that a healthy, adequately watered plant cannot have more than a cultivar-specific number of green leaves. Increases in leaf length and leaf width are derived from plant extension rate, hence allowing for the calculation of leaf area expansion, which is done for individual leaves on a tiller. Leaves stop expanding once they reach a maximum allowable blade area until a specific number of leaves have formed. The fraction of light intercepted by the crop is determined by the green leaf area index, calculated as the product of the mean area of green leaves per tiller and the tiller population. The interception of photosynthetically-active radiation by the canopy is calculated by Beer’s law, using a variable extinction coefficient during the crop development. Stalk water content Stalk water is assumed to depend on both on the container volume (a positive effect), which can be represented by the dry weight of the structural material, and the quantity of sucrose stored (a negative effect). Water stress The model assumes water stress regulating different plant processes (carbon assimilation, expansive growth) by two water stress factors. These stress factors, which represent the rate of the different processes (i) relative to the unstressed rate, are determined by potential water supply and demand using the approach of the CERES model: The thermal time taken for full recovery is controlled by a species parameter. Potential demand is calculated from the atmospheric evaporative demand (based on the FAO-56 reference evaporation) modified by two crop coefficients. Actual water uptake is calculated in the DSSAT water balance module following the CERES maize approach. A species parameter limits daily water transport to a maximum value of $0.07 \text{ cm}^3 \text{ cm}^{-1}$. Nitrogen stress It is not modeled in the used DSSAT/CANEGRO version.	Leaf appearance rates declines as a continuous function of cumulative thermal time. The faster early expansion of LAI in ratoon crops is simulated via a shorter lag time for regrowth and by tillering making larger contribution to leaf area development in a ratoon. The daily rate of senescence of green leaf area is calculated as the maximum of four rates: aging, light competition, water stress and frost. Aging causes senescence by not allowing at any time more than a certain number of fully-expanded green leaves per stalk. Light competition is simulated to induce senescence once fractional radiation interception reaches 0.85. Green leaf area index is the product of green leaf area per stalk and the number of stalks per unit ground area. Green leaf area per stalk is simulated by summing the fully-expanded area of successive leaves that appear on each stalk, and adding a correction factor for the area of expanding leaves. Profiles of leaf area per leaf are input as genotype coefficients. LAI is used in the model to intercept incident solar radiation following Beer’s Law, using a constant radiation extinction coefficient. Stalk water is calculated based on the assumption of both crop age and stalk sugar content can independently influence stalk water content. For every gram of structural stalk stem grown, a weight of water is considered to have been accumulated by the cane stalk. This relationship varies with thermal time, ranging from $10 \text{ g water g}^{-1}$ dry matter initially, to $4.5 \text{ g water g}^{-1}$ dry matter late in the crop life cycle. Water stress is assumed to reduce the rate of leaf area expansion and radiation-use efficiency, following the concepts embodied in the CERES models. Water stress also anticipates the sucrose accumulation and increases sucrose concentration. Transpiration demand is modeled as a function of the current day’s crop growth rate, divided by the transpiration-use efficiency. When soil water supply exceeds transpiration demand, assimilate fixation is a function of radiation interception and radiation use efficiency. When soil water supply is less than transpiration demand, assimilate fixation is a function of water supply and transpiration efficiency and the vapor pressure deficit. The potential rate of extraction in a layer is calculated using a rate constant, which defines the fraction of available water able to be extracted per day. Crop nitrogen demand is simulated using an approach similar to that used in the CERES models. Crop N demand is calculated as the product of maximum tissue N concentration and the increment in tissue weight. Separate N pools are described for green leaf, cabbage, stalk and dead leaf. Only the leaf N concentrations influences crop growth processes. Growth is unaffected until leaf N concentrations fall below a critical concentration. Nitrogen stress is proportional to the extent to which leaf N falls between the critical and the minimum N concentration. Senescing leaves (and the associated leaf sheaths contained in the cabbage pool) are assumed to die at their minimum N concentrations and the balance of the N in these tissues is translocated to the green leaf and cabbage pools. N uptake cannot exceed N demand by the crop and is simulated to take place by mass flow in the water that is used for transpiration. Nitrogen stress also influences biomass partitioning in the stem, in a similar fashion to that described above for water stress.

temperatures above this slowing the accumulation of thermal time. In contrast, phenology in the DC model uses two upper cardinal temperatures.

União (Site 1) was the warmest site among those evaluated (Table 1), and it was where we found the largest difference (even for the irrigated treatment) in SFM between the two models (Fig. 2a and b). The models also had differences in simulated LAI at this site (Fig. 3a and b). Phenology was also affected, with the DC model showing the fastest emergence at nine days compared to AS (Figs. 2a, b, and 3a, b). The earlier plant emergence in the DC model compared to the AS at Site 1 mostly explains the lower biomass accumulated for AS compared to DC (Fig. 2a and b). Based on these observations, it we proposed that the performance of the AS model could be improved under hot climates by changing the algorithm for degree-day integration to use two optimum temperatures (i.e., a “trapezoidal” function) instead of having only one optimum temperature (a “triangular function”).

NCo376 parameters overestimate tiller number for DC (Fig. 4a) but underestimated them for AS (Fig. 5a) at 500–2000 °C d, and that way AS underestimated LAI. DC model showed a delay in the canopy development up to 1800 °C d even after the calibration (Fig. 4a and b). Calibration process little changed the GLAI values in AS model, which were almost stable and close to $8 \text{ m}^2 \text{ m}^{-2}$ after 2000 °C d (Fig. 5a and b). DC simulated GLAI around $5 \text{ m}^2 \text{ m}^{-2}$ after 2000 °C d (Fig. 4b), which was close to the observed values.

3.2. Model intercomparison after calibrating for the RB867515 cultivar

After calibrations were performed using the experimental data from the seven experiments throughout Brazil, both of the crop models proved capable of simulating crop phenology, total above-ground biomass, leaf area, and stalk fresh mass in daily time steps for sugarcane grown in Brazil (Figs. 6 and 7). The models and

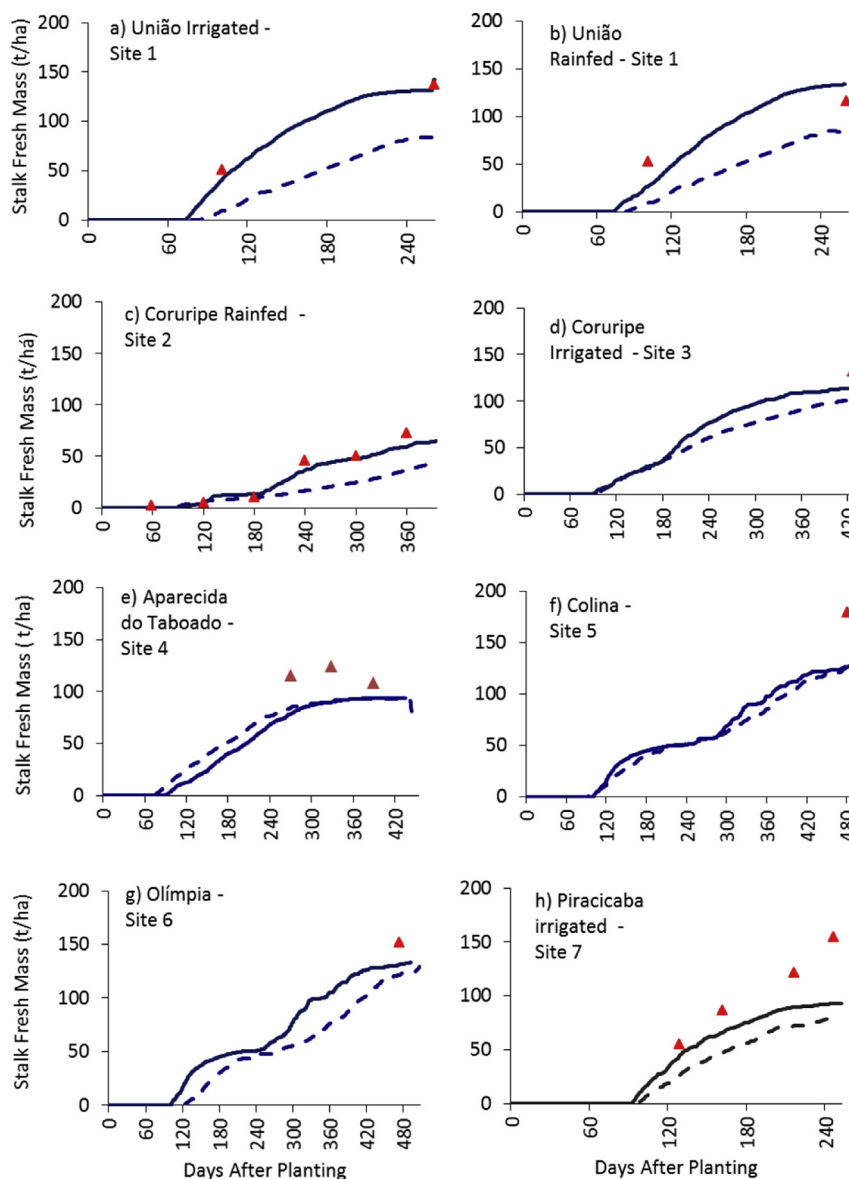


Fig. 2. Simulated stalk fresh mass (t ha^{-1}) by DSSAT/CANEGRO (solid line) and APSIM-Sugarcane (dashed line) using the cultivar parameters for NCo376 for different environments around Brazil. Red triangles are the experimental data for RB867515 observed in each Site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

calibration features may be grouped according to the major processes of phenology, tillering, leaf area dynamics, radiation use efficiency and soil water dynamics.

To represent the differing effects of temperature and radiation on leaf expansion at different growth stages, both models simulated the leaf area dynamics dependent on the crop phenological stage. The leaf area development and expansion of the LAI were simulated through the lag time for growth (or regrowth in the case of ratoon crops) of shoots after planting (or harvest). In both models, this lag time can be adjusted through the cultivar-specific parameters (Table 2).

In the DC model leaf area is related to tiller number, which is in turn regulated by temperature and the genetic parameters that allow adjustments for stalk emergence, start of tillering, peak tillering, and tiller number at maturity (Table 3). The main difference between the South African NCo376 and Brazilian RB867515 cultivars was the value of the CHUPIBASE parameter (Table 3), which reduces the thermal time required for stalk emergence. In

AS, the *tt_emerg_to_begcane* parameters handles the tillering initiation and it was reduced from 1800 °C d (for NCo376) to 650 °C d for RB867515 cultivar.

As described above (Table 2), in the AS model the additional leaf area associated with tillers that appeared and subsequently died was captured through a calibrated tillering parameter (Table 2). The set of values of the tillering parameter (*tillerf_leaf_size*) for RB867515 is showed Table 4.

For the AS model, we set the maximum green leaves for RB867515 at 13 to improve the model performance. In contrast, for the DC model we set a maximum of 10 green leaves (Table 3), a value which was close to that observed in field experiments (Nassif et al., 2012; Marin et al., 2012).

To simulate solar radiation utilization, both of the models applied an approach based on the radiation use efficiency (Table 2). The DC model RUE is a cultivar specific parameter that represents the units of dry mass of assimilate produced before respiration per unit of photosynthetically active radiation at the canopy level. The

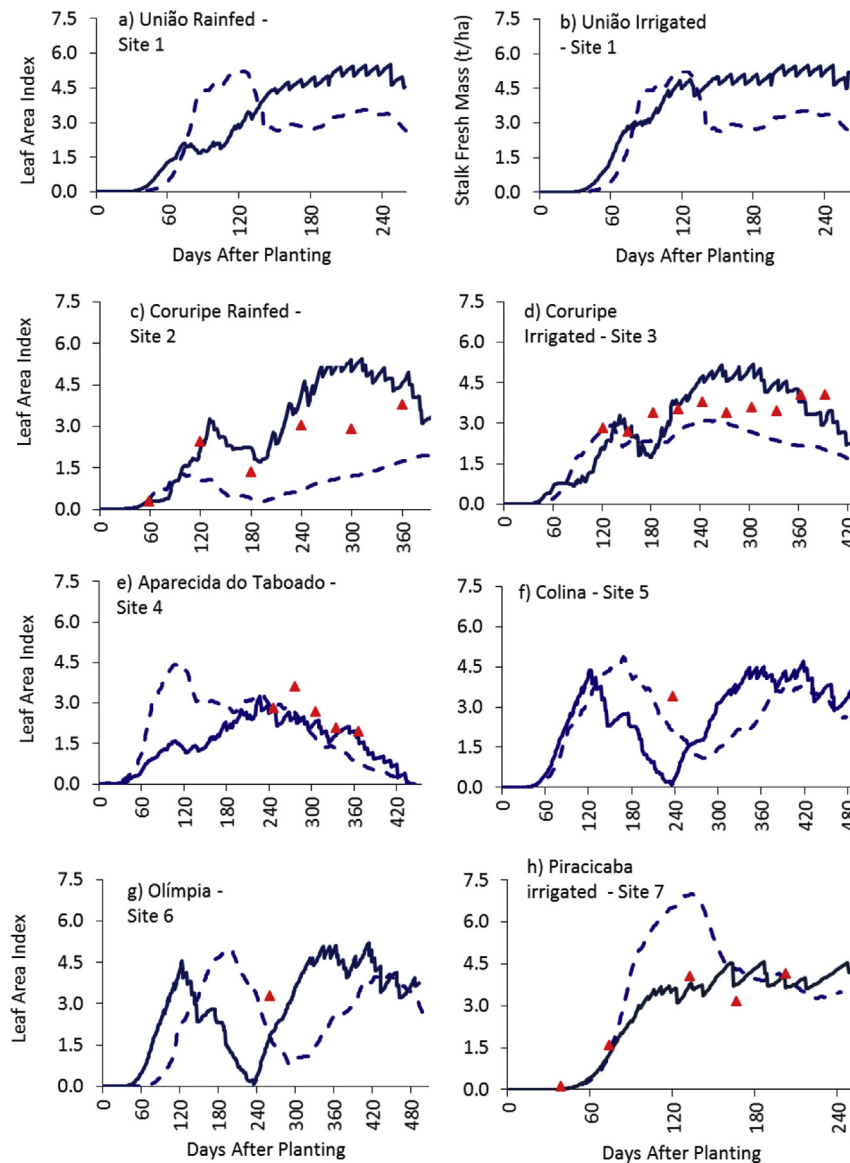


Fig. 3. Simulated leaf area index by DSSAT/CANEGRO (solid line) and APSIM-Sugarcane (dashed line) using the cultivar parameters for NCo376 for different environments around Brazil. Red triangles are the experimental data for RB867515 observed in each Site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

value RUE for NCo376 was for 9.9 g MJ^{-1} , and we used the same value for RB867515 following the philosophy (i.e. that RUE is generally stable among cultivars) adopted in the AS model. In the AS model, RUE is defined as the biomass accumulated per unit of intercepted radiation (as the process of respiration is not modeled explicitly) and the value for RUE has been found to be stable among cultivars from many countries (Muchow et al., 1994; Robertson et al., 1996; Keating et al., 1999; Cheeroo-Nayamuth et al., 2000). The RUE default value of 1.80 g MJ^{-1} for plant crops and 1.65 g MJ^{-1} for ratoon crops was retained for the AS model (Table 2, Keating et al., 1999). The applicability of this value to Brazilian sugarcane cultivars needs to be confirmed in detailed field studies like those undertaken by Muchow et al. (1994) and Robertson et al. (1996).

In the crop model calibrations, the AS model predicted slower canopy development than the DC model, and the AS model also tended to underestimate sugarcane growth at the beginning of the season (Fig. 6). However, at the end-of-cycle SFM estimated by the AS model was fairly consistent with observed values (e.g., Fig. 6a, b,

and f). In comparison, the DC model simulated earlier plant emergence, faster crop development and higher SFM estimates. The correlation coefficients of the model estimates to the observed SFM were similar for both models, although the DC model had a lower RMSE for the SFM and a slightly higher RMSE for the LAI (Table 5).

Under moderate water stress events, the DC model restricted the canopy development earlier and more severely than the AS model, which resulted in a higher variability in predictions for the LAI over time in the DC simulations and showed that the DC model's leaf area algorithm was more sensitive to water stress than that of the AS model (e.g., Fig. 7c, e and f). Severe water stress events were estimated well by both models (Fig. 7c); however, the AS simulations had less variability in the LAI (Fig. 7). On average, the DC model simulated lower LAI maximum values than the AS model.

The experiments at Coruripe (Fig. 7a and b) also highlighted the model differences regarding the effects of water stress on the LAI. In the rainfed treatment (Fig. 7a), LAI predicted by the AS model was lower than the measured data for almost the entire cycle and

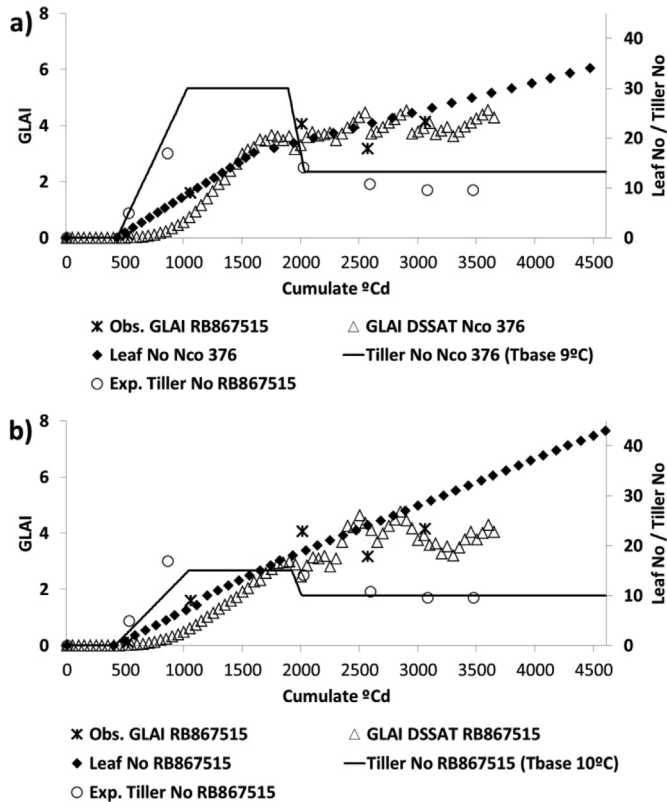


Fig. 4. Schematic representation of DSSAT/CANEGRO tiller number (tiller no.), leaf number (leaf no.), and green leaf area index (GLAI, $\text{m}^2 \text{m}^{-2}$) for Site 7 using 30 year weather data from Piracicaba with (a) the parameters for NCo376 and (b) parameters calibrated against data for RB867515. Observed GLAI and tiller number for RB867515 are also shown.

showed a severe decline due to a stress period approximately 180 days after planting (DAP). The DC model simulated a faster LAI increase of up to 120 days, followed by a strong decrease to approximately 180 DAP. Subsequently, the DC model simulated the LAI increasing from 1.6 to 5.0 in 50 days, whereas the AS model simulated the LAI increasing from 0.2 to 2 in 160 days. Both models simulated higher LAI values than rainfed treatment (Fig. 7a) than the partially irrigated treatment (Fig. 7b), with the AS model producing more stable LAI through the crop cycle and the DC model simulating higher LAI variability. After 260 DAP (Fig. 7b), the AS model decreased the LAI from 3.1 to 1.6; however, the DC model delayed the decrease by approximately 300 DAP, with the LAI reducing from 5.7 to 2.5. These differences showed that the DC model was more sensitive to water stress, and they also indicated differences in the leaf senescence process (Table 2).

3.3. Performance of single- and multi-model simulations

After the calibration process, the AS and DC models exhibited a similar ability ($r = 0.94$ and 0.95 , respectively) to predict sugarcane biomass in the field in Brazil (Table 5) based on the same data used in the calibration. The RSME values found in this study were generally comparable to those from previous studies (Keating et al., 1999; O'Leary, 2000; Cheeroo-Nayamuth et al., 2000; Singels and Bezuidenhout, 2002; Marin et al., 2011). The average results from both models (hereafter referred to as the 'Mean') produced better estimates of the SFM and LAI than the individual model estimations throughout the sites and seasons (Fig. 8, Table 5). This result was particularly true for the LAI predictions and consistent with the

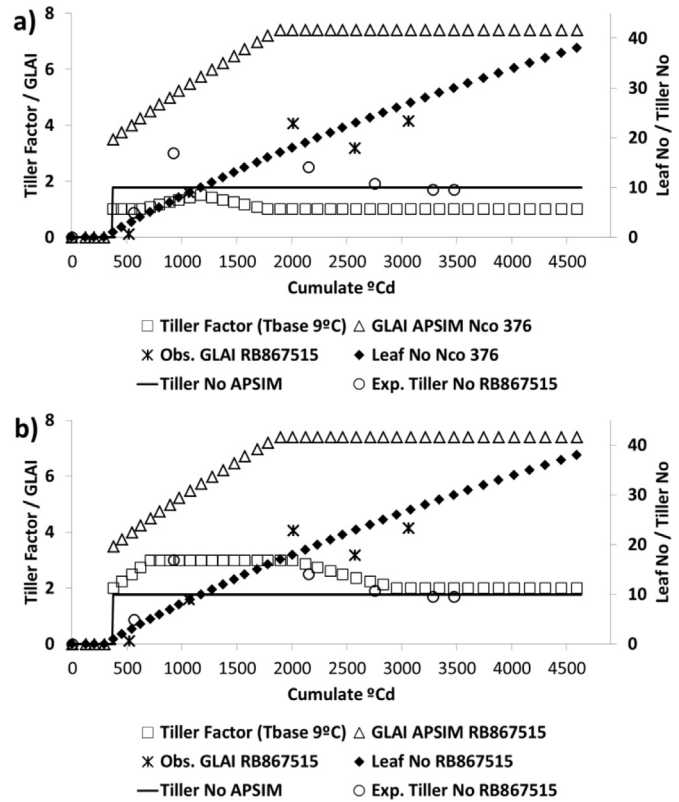


Fig. 5. Schematic representation of APSIM-Sugar tiller factor, leaf number (leaf no.), and leaf area index (GLAI, $\text{m}^2 \text{m}^{-2}$) simulated for Site 7 using 30 year weather data from Piracicaba with (a) the parameters NCo376 and (b) parameters calibrated against data for RB867515. Observed GLAI and tiller number for RB867515 are also shown.

hypothesis of Asseng et al. (2013) and Martre et al. (2014) that improved performance of the mean or median values from multi-model simulations compared with single-model simulations. The simulation errors from the 'Mean' model were generally lower than those found by Keating et al. (1999), O'Leary (2000), Cheeroo-Nayamuth et al. (2000), Singels and Bezuidenhout (2002) and Marin et al. (2011). Thus, applying the two models (in their current form) is likely to give the more accurate predictions than focusing on one model alone.

Van Oijen and Ewert (1999) and Walker et al. (2003) indicated that there were three major sources of simulation uncertainty: (i) input data, (ii) parameterization and (iii) model structure. The narrow range of variation of the SFM simulations (Fig. 8) indicated a lower uncertainty related to the SFM estimates than that found for the LAI, and the inadequate LAI simulations might have been related to the third uncertainty source noted above, model structure. The models used daily weather variables as the input data in addition to the soil physical properties, initial conditions and basic crop and soil management information. The uncertainties related to the input data were important for several sites where on-site weather measurements or rough soil descriptions were not obtained; this was mainly occurred for the sites in southern Brazil. Nevertheless, even the experiments under more detailed control had measurement errors and small-scale variability in annual rainfall that were related to variables such as the rainfall measurements that used standard rain-gauges, which usually underestimated the ground precipitation (Legates and Willmott, 1990; Subedi and Fullen, 2009). In our study, almost all the series of weather data had missing values that increased the simulations' uncertainty and required interpolated estimates or data from nearby stations.

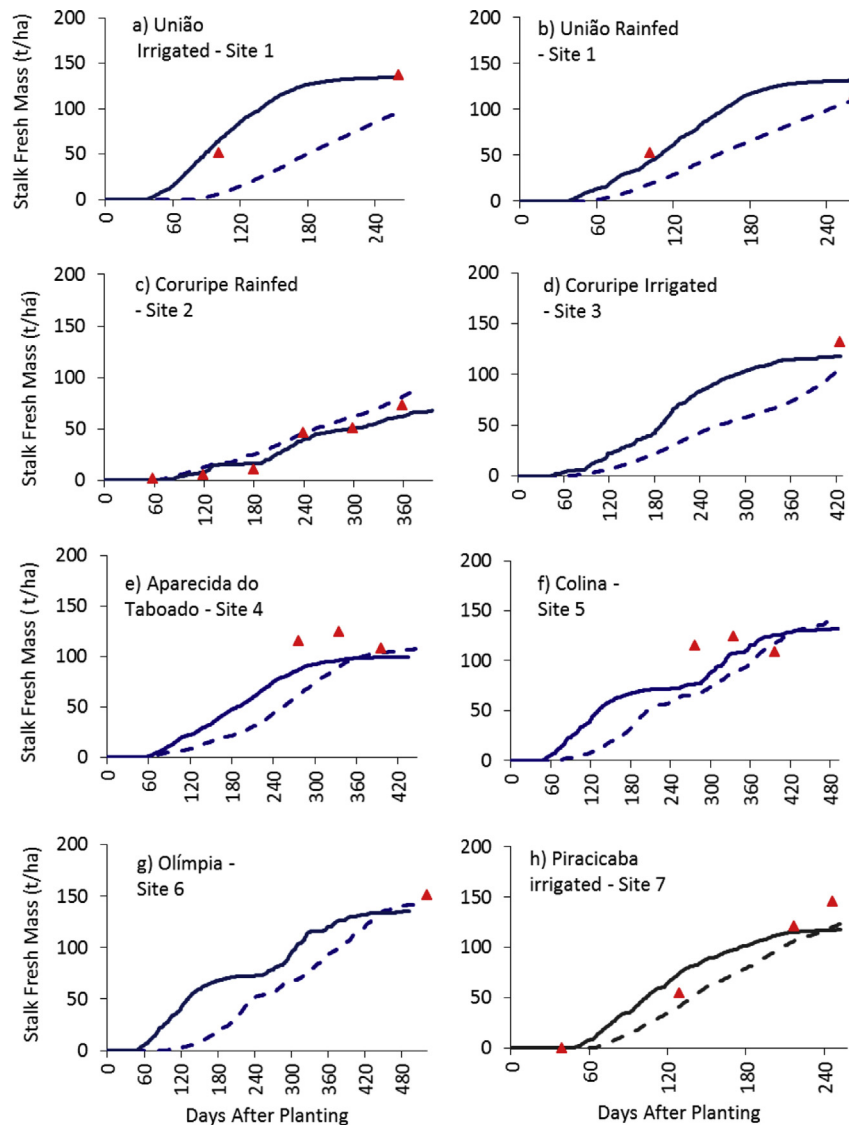


Fig. 6. Simulated stalk fresh mass (t ha^{-1}) by DSSAT/CANEGRO (solid line) and APSIM-Sugarcane (dashed line) after calibration for RB86-7515 for different environments around Brazil. Red triangles are the experimental data for RB867515 observed in each Site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

We calibrated only the crop phenology and canopy-related parameters based on reasonable quality observations, which were only obtained from Sites 3 and 7 because they had detailed and regular crop measurements throughout the cycle. Although we used the same data for both models (Figs. 6 and 7), there were differences in how we interpreted and translated the models. Following Palosuo et al. (2011), the additional cultivar-specific parameters that were required for the models were the default model values and data from the literature and/or an earlier application of the models; the quality of the data was dependent on its applicability (geographic distance, same/similar varieties, etc.) and the quality of the initial empirical data source. By definition, the crop simulation models were simplifications of reality; some of the simulations included oversimplifications that led to marked discrepancies between the simulated and measured data. For both models analyzed here, we observed that the main differences were in the sugarcane tillering, canopy and leaf area development algorithms. These differences were reflected to the differences in the SFM simulations, which were primarily related to the models'

response to temperature in terms of phenology and limited the AS model's ability to simulate crop growth under hot climates; the SFM simulations were also related to the canopy structure, which apparently underestimated the biomass production rates in both models.

3.4. Sensitivity analysis for simulating the climate change scenarios

3.4.1. Effect of atmospheric carbon dioxide concentration

The AS model simulated SFM increases with an average of +4% per 100 ppm (Fig. 9a). The CO_2 responses in the AS model resulted from changes in the RUE and water use efficiency, which reduced the water stress (by increasing the crop resistance for water vapor diffusion) at the same time slight increased the gross photosynthetic rate. The DC model simulated lower crop responses regarding CO_2 compared to AS model. By increasing CO_2 concentration from 390 to 750 ppm, LAI increased less than 5% (Fig. 10d), while stalk fresh mass increased less than 2%.

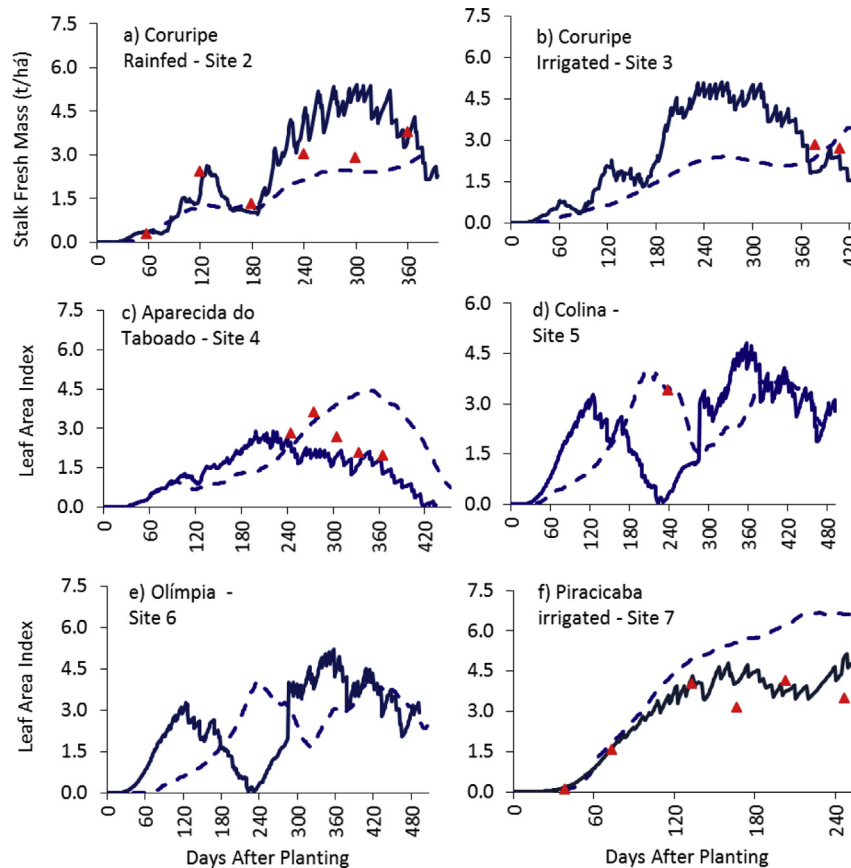


Fig. 7. Simulated leaf area index by DSSAT/CANEGRO (solid line) and APSIM-Sugarcane model (dashed line) after calibration for RB86-7515 for different environments around Brazil. Red triangles are the experimental data for RB867515 observed in each Site. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In our implementation of the AS model, elevated concentrations of atmospheric CO_2 were represented by multiplying the default transpiration efficiency and radiation use efficiency coefficients by the modifiers for the effect of CO_2 concentration $[\text{CO}_2]$, as follows

Table 3

Parameters developed for the cultivar RB867515 for the DSSAT/CANEGRO model. Parameter values for cultivar NCo376 previously developed for DSSAT/CANEGRO are shown for comparison.

Parameter	Description	Units	RB867515	NCo376
PARCEMAX	Radiation use conversion efficiency	g MJ^{-1}	9.9	9.9
APFMX	Aboveground biomass partition fraction of a mature crop	Fraction	0.88	0.88
STKPFMAX	Stalk partition fraction of aboveground biomass	Fraction	0.65	0.65
LFMAX	Maximum green leaf number	Leaves stalk $^{-1}$	10.	12.
MXLFAREA	Area of the biggest leaf	cm^2	594	360
MXLFARNO	Leaf no. of the biggest leaf		18	14
PI1	Phyllochron for leaves younger than leaf no. = Pswitch	$^{\circ}\text{C d}$	89	69
PI2	Phyllochron for leaves older than leaf no. = Pswitch	$^{\circ}\text{C d}$	107	169
PSWITCH	Leaf no. where switch in phyllochron occurs		18	18
TTPLNTEM	Thermal time requirement for emergence of a plant crop	$^{\circ}\text{C d}$	408	428
CHUPIBASE	Thermal time requirement for start of stalk elongation	$^{\circ}\text{C d}$	547	1050
TT_POPGROWTH	Thermal time requirement to peak tiller population	$^{\circ}\text{C d}$	628	600
MAX_POP	Peak tiller population	m^{-2}	16	30
POPTT16	Final tiller population	m^{-2}	10	13.3

below (Howden et al., 1999; Webster et al., 2009; Biggs et al., in press):

$$\text{rue_co2_modifier} = (1/350 \times 0.05) \times [\text{CO}_2] + 0.95 \quad (1)$$

$$\text{tue_co2_modifier} = 0.0008 \times [\text{CO}_2] = (1 - 0.0008 \times 350) \quad (2)$$

The DC model simulated a small direct response to photosynthesis (Fig. 11); however, only transpiration responded to CO_2 concentrations by increasing the leaf resistance to water vapor loss, which led to a higher water-use efficiency and reduction of water stress. The algorithm for photosynthesis in the DC model calculated the daily increments of total biomass using a radiation-use efficiency approach (Singels and Bezuidenhout, 2002) and CO_2 fertilization effect algorithm (Singels et al., 2013), which accounted for the rate of photosynthesis (P_G) and yields (Eq. (3)):

$$P_G = F_i \cdot \text{PAR} \cdot \text{RUE} \cdot R_{PG} \quad (3)$$

where PAR is photosynthetically-active solar radiation (MJ m^{-2}), RUE is radiation use efficiency (g MJ^{-1}) and F_i is fractional interception of PAR.

The standard P_G at a CO_2 level of 330 ppm was adjusted using an adjustment factor (R_{PG}) that was dependent on the amount of CO_2 . Rather than defining this relationship mathematically, the DSSAT provided a mechanism for interpolating the R_{PG} from a set of coordinate pairs (Gerrit Hoogenboom, pers. comm., 2011). The data points were defined in the species file, and the function for maize (from the 2011 DSSAT v.4.5 source code) was used for sugarcane.

Table 4
Parameters developed for the cultivar RB867515 for the APSIM model. Parameter values for cultivar NCo376 previously developed for APSIM are shown for comparison.

Parameter	Description	Units	RB867515				NCo376			
leaf_size	Area of the leaves	mm ²	3500	45,000	59,400	59,400	3000	7000	12,000	28,000
leaf_size_no	Leaf number		1	5	18	26	1	4	14	20
cane_fraction	Stalk partition fraction of aboveground biomass	Fraction	0.70				0.65			
sucrose_fraction_stalk	Sucrose fraction in the stalk	Fraction	1	0.55			1	0.55		
stress_factor_stalk	Stress factor on stalk growth		0.2	1			0.2	1		
tt_emerg_to_begcane	Thermal time emergence to beginning cane	°C d	650				1800			
tt_begcane_to_flowering	Thermal time beginning cane to flowering	°C d	8000				6000			
tt_flowering_to_crop_end	Thermal time beginning cane to flowering	°C d	2000				2000			
green_leaf_no	Maximum green leaf number	Leaves stalk ⁻¹	10				10			
tillerf_leaf_size	Tiller leaf size factor		2	3	3	2	2	1	1	1.5
tillerf_leaf_size_no	Tiller no.		1	5	10	18	26	1	4	10
										16
										26

In addition to its effect on photosynthesis, the DC model also simulated the impact of CO₂ on the stomatal resistance and transpiration (e.g., Long et al., 2004) following the method proposed by Allen et al. (1985) in Eq. (2):

$$r_s = \frac{(3.28 \cdot 10^{-2} - 5.49 \cdot 10^{-5} \cdot [\text{CO}_2] + 2.96 \cdot 10^{-8} \cdot [\text{CO}_2]^2)^{-1} + r_{lb}}{\text{LAI}} \quad (4)$$

where r_s is the canopy resistance to water vapor diffusion (s m⁻¹), [CO₂] is the [CO₂] expressed in μmol mol⁻¹, r_{lb} is leaf boundary resistance (assumed 10 s m⁻¹) and LAI is leaf area index.

Eq. (4) predicted stomatal resistances of 62 and 135 s m⁻¹ at a CO₂ level of 380 and 880 μmol mol⁻¹, respectively. Although the values were derived for sweet corn, they were fairly consistent with the sugarcane stomatal diffusion resistances measured under ambient atmospheric CO₂ by Venkataramana et al. (1986), Grantz and Meinzer (1990), Souza et al. (2008), Vu and Allen (2009), and Nassif et al. (2013). The DS model used an adjustment factor for potential transpiration that was derived from the theoretical ratio of the grass reference evapotranspiration calculated by the Penman–Monteith method, and the canopy resistance was calculated for the relevant CO₂ based on the reference CO₂ level of 330 μmol mol⁻¹ (Marin et al., 2012). The reference level of CO₂ was derived from the C4 plants that have a CO₂ concentrating mechanism in the leaf mesophyll cells that binds CO₂ and transports it as a four-carbon molecule to the vascular bundle sheath cells for decarboxylation (Allen et al., 1985). It is evident that more research is needed to improve this algorithms considering the few

experimental data used, that none of them was collected on sugarcane experiments, and even the few sugarcane experiments carried out around the world were not considered in the development of algorithms for both models.

3.4.2. Effect of changes in rainfall

Both models simulated positive SFM and LAI responses to increased rainfall (+30%) for such environment (Figs. 9b and 10b), with major responses [+12% in SFM and +8% for LAI (Fig. 9e)] produced by AS. DC simulated smaller increases in SFM and LAI (Fig. 10b and d), and this result is in part caused by quicker root development in a deep soil profile (4.5 m). Thus, roots get deeper sooner than AS, ensuring water supply from the soil reservoir and reducing the influence of dry periods on crop development and growth.

Both models showed bigger sensitivities to rainfall reduction than for increases in rainfall. SFM decreased by 30% and 16% when rainfall was reduced by 30% for AS and DC, respectively (Figs. 9b and 10b). This was consequence of reasonably adequate rainfall amount in Piracicaba (in average 1253 mm year⁻¹) for sugarcane growth, as well as because rainfall variation were done by changing rainfall amounts to rainy days while maintaining the same rainfall distribution overtime. Therefore, dry days remained without rain and periods under water stress have changed little despite the increased amount of rain for time periods when the rainfall was adequate. Although the increased rainfall has caused minor influence in averaged SFM, temporal yield variability variables was reduced as rainfall was increased as indicated by the error bars in Figs. 9b and 10b.

Both models simulated low increases of LAI as the rainfall increased (Figs. 9e and 10e). However, both showed higher sensitivity of LAI to the reduction of rainfall, being AS more sensible than DC model. The simulated LAI was reduced by nearly 50% and 15% for AS and DC, respectively. This was caused by a decrease in number of green leaves in both models together with higher crop water stress simulated by AS.

3.4.3. Temperature effects

The AS model simulated a marked decrease in SFM and LAI with increasing temperatures at all of the levels (Fig. 9c and f). In contrast, the DC model simulated the maintenance of SFM up to +3 °C followed by a decrease at +6 °C and +9 °C (Fig. 10c). DC also simulated a continuous and slight decrease in LAI for the all range of temperature increase (Fig. 10f). The temperature range of 12 °C led to a range in SFM of 46% in the AS model. The corresponding range in yields from the DC model was 15%. These

Table 5
Statistical agreement of APSIM-Sugarcane and DSSAT/CANEGRO simulated values of stalk fresh mass (SFM, t ha⁻¹) and leaf area index (LAI) to observed values.

	Agreement measures	SFM	LAI
APSIM-Sugar	RMSE	20.09	1.35
	r	0.94	0.45
	Mean obs.	90.60	3.39
	Mean sim.	78.81	2.72
DSSAT/CANEGRO	RMSE	18.27	1.10
	r	0.95	0.67
	Mean obs.	90.60	3.39
	Mean sim.	82.74	2.83
Mean	RMSE	18.01	0.97
	r	0.97	0.66
	Mean obs.	90.60	3.39
	Mean sim.	80.31	2.78

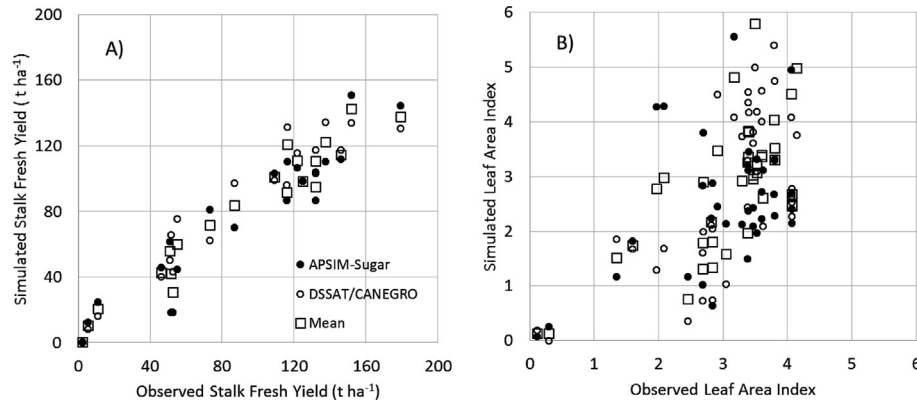


Fig. 8. Simulated and observed stalk fresh mass (A) and LAI (B) for 8 experiments around Brazil for DSSAT/CANEGRO and APSIM and multi-model mean.

differences in model responses to temperature must be evaluated against accurate temperature transect data, which may be difficult to find. However, limited temperature-gradient greenhouse data (Allen et al., 1985) suggested that sugarcane biomass increased at temperatures above ambient.

Rates of photosynthesis, respiration, phenology, expansive growth and evapotranspiration are influenced by air temperature in the DC model (Table 2), while AS only consider the influence of temperature on RUE (and not directly on photosynthesis and respiration rates) below 15 °C and above 35 °C, which seldom occur in Piracicaba. AS model simulates the nitrogen cycle influenced by temperature, and the nitrogen stress is quantified using an index

(nfact) ranging from 1 (fully stressed) to zero (no stress). The same procedure is used for accounting the water stress (swdef) by using the Ritchie approach (Ritchie, 1998, Table 2) for computing separately the stress effect on expansive growth and photosynthesis. For AS, nfact increased by 1.2% for the +3 °C scenario, while swdef increased by 7.7% and 30.1% for the +3 °C and +6 °C scenarios, respectively, as a consequence of the increase of potential evapotranspiration (+8.2% in relation to current scenario for +3 °C). In general, water stress simulated by AS are usually greater than those produced by DC. Adding to this the fact that RUE is not usually affected by air temperature in Piracicaba, and this explains the SFM rise when air temperature is reduced by −3 °C (Fig. 9c).

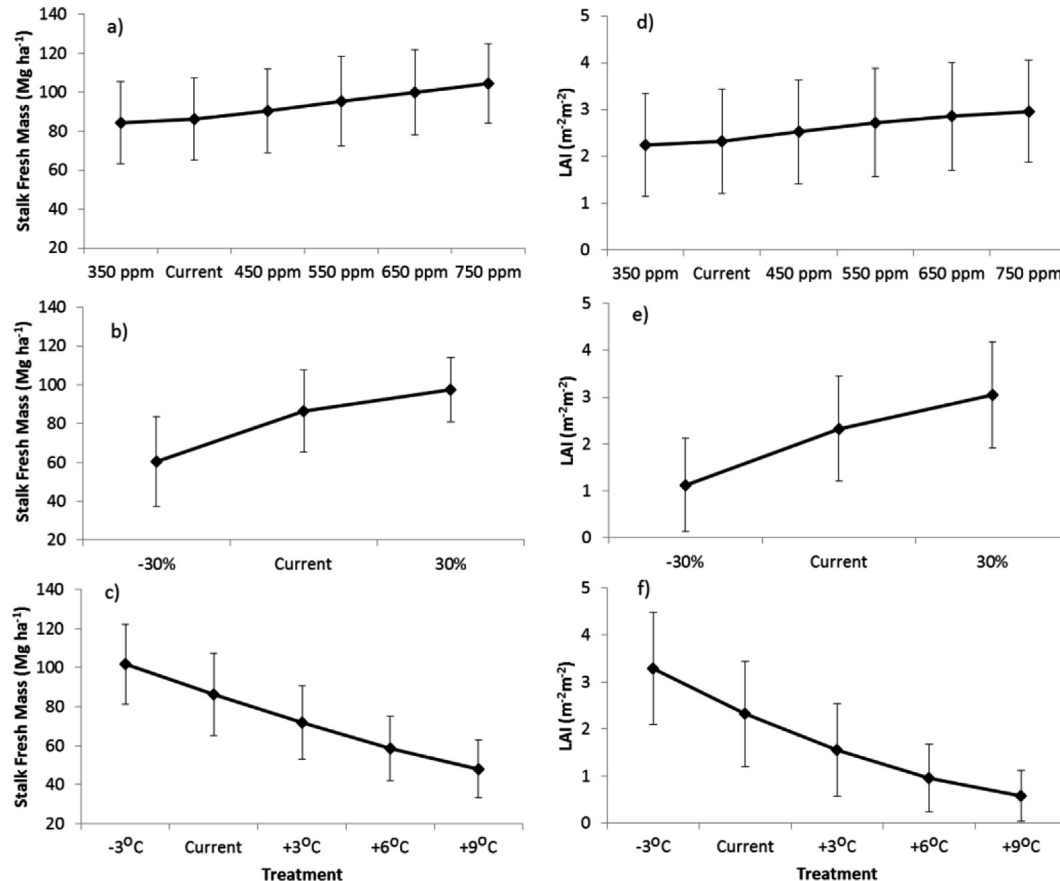


Fig. 9. APSIM-Sugar sensitivity for stalk fresh mass and leaf area index to CO₂, rainfall and air temperature for 30-year weather series of Piracicaba-SP.

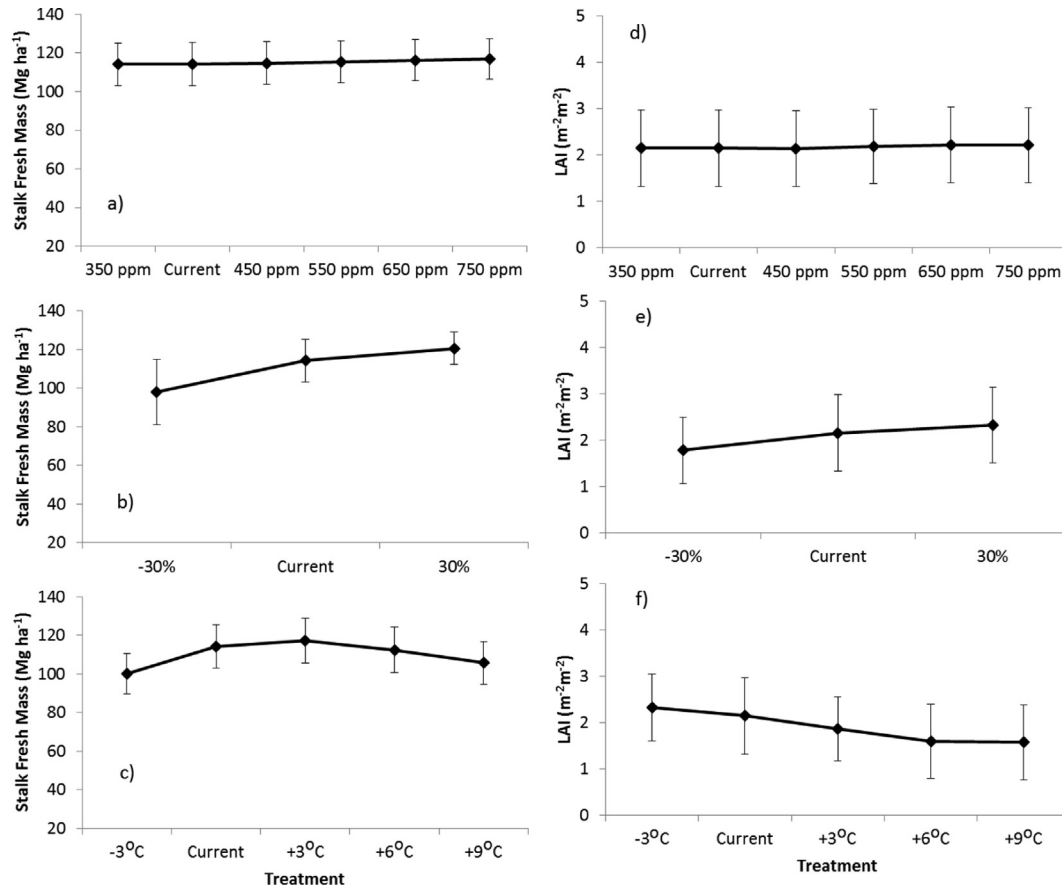


Fig 10. DSSAT/CANEGRO sensitivity for stalk fresh mass and leaf area index to CO₂, rainfall and air temperature for 30-year weather series of Piracicaba-SP.

DC does not simulate the nitrogen stress and the water stress is quantified using a soil water deficit factor (SWDF1, see Singels et al., 2008) that ranges from 1 (no stress) to zero (fully stressed). The average SWDF1 value increased by 6.7 and 26.7% for the +3 °C and +6 °C scenarios. Increased temperatures caused large increase in potential evapotranspiration (+7.8% in relation to the current scenario) for +3 °C rise and accelerated canopy development (80% canopy cover reached 15 days sooner for +3 °C). Instead, for -3 °C, in average, there was a delay of 14 days in the tillering peak. This led

to an increase in canopy photosynthesis and actual crop evapotranspiration 6.6 for +3 °C due to increased interception of radiation. The increased evapotranspiration led to an increase in severity of water stress.

4. Conclusion

Despite calibration to some of the best data on sugarcane growth and development available on the widely grown cultivar RB867515 in Brazil, the two most widely used sugarcane models in the world (APSIM-Sugar and DSSAT/CANEGRO) did not necessarily accurately predict canopy and stalk development through time. Both models underestimated the leaf area index, canopy dimension for intercepting solar radiation, and stalk fresh mass for cultivar RB867515. AS model under predicted yield at the hotter sites, likely because the algorithm for computing the degree-days is based in only one upper cardinal temperature while DC uses two upper temperatures.

The two models had different sensitivities to changes in temperature, rainfall and atmospheric CO₂ concentrations likely to occur under climate change. Both models simulated increase sugarcane growth in response to higher rainfall and CO₂ concentrations, and lower growth for +6 °C and +9 °C scenarios. However, the extent of these changes was different in the two models. The AS model exhibited a higher sensitivity to the reduction of rainfall than the DC model. AS also simulated a decrease in leaf area index with increasing temperature at all of the levels, whereas the DC model simulated a similar SFM up to +3 °C. The differing predictions of sugarcane sensitivity to changed climatic conditions suggests that further studies are needed on the physiology of sugarcane at higher

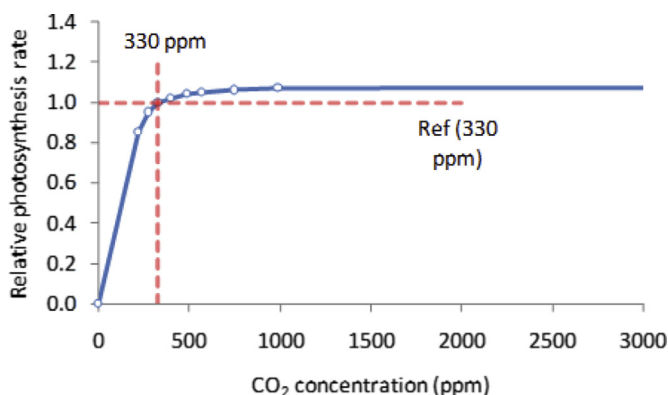


Fig. 11. Relative variation of photosynthesis rate function of atmospheric carbon dioxide concentration, for the DSSAT/CERES-Maize model. The dashed lines indicate the reference relative photosynthesis rate of one at a value of 330 ppm CO₂. Circles represent the data points that define the function (Gerrit Hoogenboom, 2011, pers. comm.).

temperatures and CO₂ concentrations to reduce uncertainty about the potential effects of projected climate change.

The mean of simulations from both models produced better estimations of the experimental data on stalk fresh mass and leaf area index than predictions from either model individually for all of the sites and seasons. Thus, applying the two models (in their current form) is likely to give the more accurate predictions than focusing on one model alone. Given the different sensitivities to climate change exhibited by the two models, using the mean of simulations from both models may give more certain predictions of climate change impacts for sugarcane than results from either model.

While best data on sugarcane growth and development in Brazil was used in this study, the experiments were not specifically designed to provide the process level data needed for robust testing and development of crop models. There is thus a need for targeted experiments to understand the physiology of the RB867515, and other cultivars so they can be better represented in sugarcane models for application in Brazil. There are a number of processes that are represented differently in the two sugarcane models included in this study. More targeted field studies might show whether the structure of one of the models is currently better suited for representing Brazilian cultivars and conditions. More generally, joint application of the these models, and some of the lesser used sugarcane models across a wide range of soils, climates and sugarcane cultivars has the potential facilitate development of better capability for simulating sugarcane.

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